

# Sporting Ethos — Implementation Plan

**GigPoint Hackathon 2026 · Track 4 (Healthcare Tech) · Milestone 1 Project:** Real-Time Patient Check-In, Live Queue, Ambient Charting & Pharmacy

## 1. Feature scope & status (all built & validated live on Supabase)

### Core — check-in confirmation (the brief)

- 1.  Universal QR check-in; **sequential auto appointment IDs** (APT-0001...); no manual ID entry
- 2.  Sub-second live queue (Supabase Realtime)
- 3.  Multi-sensory arrival alert — highlight + chime + voice ("<name> has arrived"), **sound always on**
- 4.  Queue order + position + live auto-reorder
- 5.  Branded, verifiable receipt (logo + timestamp + UUID + SHA-256 hash)
- 6.  Live metrics (measured confirm latency, waiting, LWBS, no-show)
- 7.  Status flow: waiting → in\_consult → done, plus left / no\_show / paused

**Queue intelligence** 8.  Emergency priority → jumps queue + recalculates every position/ETA on all phones 9.  Call Next → patient's phone chimes + speaks "It's your turn" (in their language) 10.  Reception + Add patient (desk walk-in registration)

**Ambient AI charting** 11.  FastAPI + Groq (Whisper + Llama 3.3 70B): audio → transcript → structured notes → PDF → patient summary 12.  Expert Console ( /expert , code-gated) + two-way reception↔expert **intercom**

**Pharmacy & revenue** 13.  Pharmacy counter ( /pharmacy ): dispense by appointment ID, **auto-match Rx → inventory**, add/remove, **payment** → **bill PDF**, stock decrement 14.  Inventory management (add/edit/remove medicines, ₹ price, stock, low-stock alerts) 15.  Pharmacy revenue in Reports + Excel

**Analytics** 16.  Reports: KPIs, gender/status donuts, age bars, arrivals trend, **booking type, top reasons/conditions from consultations, pharmacy revenue**, Excel export

**Inclusion** 17.  Language switch — **English / हिन्दी / తెలుగు** (UI + voice) 18.  Accessibility mode — large text + voice-guided check-in

**Roadmap (post-hackathon):** phone-OTP patient auth · role-based RLS · multi-clinic ( clinic\_id ) · Gemini single-call charting.

## 2. Architecture decisions (locked)

| Decision         | Choice                                                         | Rationale                                          |
|------------------|----------------------------------------------------------------|----------------------------------------------------|
| App shape        | One responsive web app, role-based views + one AI microservice | No installs; one codebase                          |
| DB + realtime    | Supabase (Postgres + Realtime)                                 | Cross-device <1s with minimal server code          |
| Appointment IDs  | Postgres sequence + next_appt_id() RPC                         | Guaranteed unique, sequential                      |
| Check-in         | Universal QR, no manual ID                                     | Simplest; ID auto-assigned                         |
| Voice            | Browser SpeechSynthesis (multi-lang), always on                | Free, offline, nothing to remember                 |
| Ambient charting | FastAPI + <b>Groq</b>                                          | 100% free tier; Whisper + Llama are extremely fast |

| Decision         | Choice                                    | Rationale                                   |
|------------------|-------------------------------------------|---------------------------------------------|
| Pharmacy revenue | Bill JSON on the check-in row             | One source powers bill PDF + revenue report |
| Reports export   | SheetJS .xlsx                             | Real Excel management can use               |
| Theme            | Clean, minimalist, light                  | Clinical, trustworthy, glanceable           |
| Deploy           | Vercel (frontend) + Python host (backend) | One-click static deploy; public URL         |

### 3. Build phases (delivered)

| Phase              | Work                                                                            | Status |
|--------------------|---------------------------------------------------------------------------------|--------|
| Scaffold           | Vite + React + Tailwind; <code>checkins</code> schema; store with mock fallback | ✓      |
| Check-in           | Universal QR → mobile form → Supabase → confirmation + receipt                  | ✓      |
| Live queue         | Realtime queue, highlight + chime + voice on arrival                            | ✓      |
| Status + emergency | Call next / done / left / no-show / paused; priority + ETA                      | ✓      |
| Sequential IDs     | Postgres sequence + RPC; removed manual ID field                                | ✓      |
| Reception tools    | + Add patient; intercom to expert                                               | ✓      |
| Ambient charting   | FastAPI + Groq; record → notes → PDF → patient summary                          | ✓      |
| Expert console     | Credential gate, Now-Consulting hero, waiting queue, intercom                   | ✓      |
| Pharmacy           | Counter (match Rx → bill → payment → stock) + inventory                         | ✓      |
| Reports            | KPIs, charts, reasons, booking type, revenue, Excel                             | ✓      |
| Inclusion          | Languages (EN/HI/TE) + accessibility mode                                       | ✓      |
| Polish             | Minimalist light UI, favicon, focus states, shared primitives                   | ✓      |

### 4. Validation (tested end-to-end, live Supabase)

- ✓ Check-in → sequential ID → dashboard in **<1s**; cross-device verified.
- ✓ Status transitions update live, no reload; emergency reorders everyone.
- ✓ Ambient charting with a **real Groq key** returns genuine structured notes.
- ✓ Pharmacy: lookup by appointment ID → Rx auto-matched → payment → bill PDF → stock decremented → revenue in Reports.
- ✓ Reports filters + Excel export; language switch + accessibility voice.
- ✓ **Production build passes (0 errors).**

### 5. Milestones & timeline (per brief, all due 25/07/2026)

| Task (brief)                                          | Deliverable                             | Status |
|-------------------------------------------------------|-----------------------------------------|--------|
| Understand problem/context/users                      | Problem framing (this doc set)          | ✓      |
| Research + finalise approach                          | Multi-channel + ambient AI + pharmacy   | ✓      |
| Solution architecture, user flow, implementation plan | <b>These 3 documents (M1)</b>           | ✓      |
| Develop core features                                 | Check-in, queue, voice, receipt, status | ✓      |

| Task (brief)                    | Deliverable                               | Status |
|---------------------------------|-------------------------------------------|--------|
| Integrate + validate end-to-end | Supabase + Groq + realtime validated (M2) | ✓      |
| Functional MVP/prototype        | Full app running (M2 doc)                 | ✓      |
| Concise presentation            | Demo script + pitch (M3 doc)              | ✓      |

## 6. Risks & mitigations

| Risk                           | Mitigation                                                                 |
|--------------------------------|----------------------------------------------------------------------------|
| Wi-Fi flaky at demo            | Local mock mode (realtime across tabs)                                     |
| Browser blocks audio           | Sound auto-primers on first interaction; highlight + chime still convey    |
| Supabase free-tier pause       | One-click restore; keys wired                                              |
| Charting backend/key down      | Falls back to a labelled sample                                            |
| Phone can't reach localhost QR | Open dashboard on the LAN address before exporting the QR                  |
| Groq key exposure              | Kept server-side in <code>backend/.env</code> ; regenerate after the event |

## 7. Roles & access

| Role      | URL                    | Code                                                         |
|-----------|------------------------|--------------------------------------------------------------|
| Reception | <code>/</code>         | open                                                         |
| Expert    | <code>/expert</code>   | <code>ethos</code> (env <code>VITE_EXPERT_CODE</code> )      |
| Pharmacy  | <code>/pharmacy</code> | <code>pharmacy</code> (env <code>VITE_PHARMACY_CODE</code> ) |
| Patient   | <code>/checkin</code>  | via QR                                                       |

## 8. How to run

```
# Frontend
npm install && npm run dev      # open the Network URL (http://192.168.x.x:5173) for phones

# Ambient charting backend
cd backend && pip install -r requirements.txt
# put GROQ_API_KEY=... in backend/.env (free: https://console.groq.com/keys)
python -m uvicorn main:app --port 8000
```

Supabase is connected via `.env` ; the dashboard badge shows **"Live"**. Database objects: tables `checkins` , `medicines` ; sequence `appt_seq` ; function `next_appt_id()` .